Java Vector

Vector vs ArrayList vs LinkedList

What is Vector?

• Vector is a very old Java class, available since JDK 1.0.
• It is a class in the Java Collections Framework.
• It stores elements dynamically, just like ArrayList.
• **Key difference:** Vector is **synchronized**. All its methods (add(), remove(), get(), etc.) are thread-safe by default.
• Because of synchronization, Vector is **slower than ArrayList** in single-threaded programs.

Why is it considered legacy?

• It is considered legacy (old style) since the Java Collections Framework was introduced in Java 2 (Java 1.2).
• Newer, better practice: use ArrayList together with Collections.synchronizedList.

```
List<String> list = Collections.synchronizedList(new ArrayList<>());
```

Comparison Table

Feature	Vector	ArrayList	LinkedList
Synchronization	Yes (Thread-safe)	No	No
Performance	Slower (because of locking)	Faster	Good for sequential insertions/removals
Storage	Dynamic array	Dynamic array	Doubly Linked List
When to use	When working with multiple threads	In normal programs	When frequent insertions/deletions are needed